

Review for Quiz 1

Welcome back to CS 2100!

Prof. Rasika Bhalerao

Quiz 1 Topics

Git

- What is a repo? What is remote / local?
- `git clone <url>` copies the remote repo to your computer
- `git add <filename>` stages the changes in the file, and `git add .` stages all changes in the directory
- `git commit -m "message"` commits the staged changes
- `git push` pushes all local commits to the remote repo
- `git pull` copies all commits from the remote repo to your local repo which are not already there

Functions and documentation

- Documentation elements (description, arguments, returns, errors)
- Defining functions with types
- Conditionals and iteration
- `None` and `Optional`

Unit testing

- Using `assert` and reading its test output
- Using `pytest.raises()` and reading its test output
- Identifying test cases

We will not ask you to test functions that print things or take user input on the exam.

Git vocab

- Repository (repo): a set of code and its history
 - local: on your computer
 - remote: on another computer (like GitHub)
- Commit
 - the codebase at a given point in time (noun)
 - to add a set of changes to the repository (verb)
 - Push: to move code from a local to remote repository

Git commands

Location	Definition	git command to put code there	Postal analogy
working area	code that you are currently writing / saving in VSCode		Writing on a paper
staging area	code that is ready to be committed	<code>git add <filename></code> or <code>git add .</code>	Add a stamp and put it in your backpack
local repository	code that has been committed	<code>git commit -m "description"</code>	Put all stamped cards in the mailbox
remote repository	code on GitHub	<code>git push</code>	Workers move cards to destinations
	copy repo from remote to local	<code>git clone <url></code>	Buying special "letter-writing paper"
	copy all commits from remote to local which are not already there	<code>git pull</code>	Getting a copy of anything anyone else sent to your destination

Poll: Which command do we use to "download" a repo from GitHub to our laptop?

1. `git clone`

2. `git add`

3. `git commit`

4. `git push`

5. `git status`

Poll: Which command do we use to "stage" changes to be committed?

1. `git clone`
2. `git add`
3. `git commit`
4. `git push`
5. `git status`

Poll: If our local repo is ahead of the remote repo by a commit, what should we use to catch up the remote repo?

1. `git clone`

2. `git add`

3. `git commit`

4. `git push`

5. `git status`

Documentation

1. Overall description of the function
2. Arguments: type and description of each argument
3. Return: type and description of what is returned
 - i. Include conditions for it to return `None`
4. Errors raised: type of error and condition to raise it

Poll: What is missing from this documentation?

```
def sarcasm(phrase: str) -> Optional[str]:
    """Returns the sarcastic version of the provided phrase, where a random
    half of the characters are uppercase, and the others are lowercase.

    Parameters
    -----
    phrase : str
        The phrase to turn sarcastic

    Returns
    -----
    str
        The sarcastic version of the phrase

    Raises
    -----
    ValueError: if phrase is empty

    """
    if phrase == '':
        raise ValueError()

    sarcastic_phrase = ''
    for character in phrase:
        if not character.isalpha():
            return None
        if random() < 0.5:
            sarcastic_phrase += character.upper()
        else:
            sarcastic_phrase += character.lower()
    return sarcastic_phrase
```

Unit Testing

assert

`assert` something that is either `True` or `False`

```
assert 2 + 2 == 4
```

pytest.raises()

Argument: error type, Indented section: function call

```
with pytest.raises(ValueError):  
    get_area_of_rectangle(-1, 4)
```

Practice Quiz 1

Practice Quiz 1

Solution to Practice Quiz 1

If you use [2026 Spring's Practice Quiz 1](#), please keep in mind that we used `unittest` instead of `pytest` for testing, so the format may look different for the third question.

Poll:

- 1. What is your main takeaway from today?**
- 2. What would you like to revisit next time?**